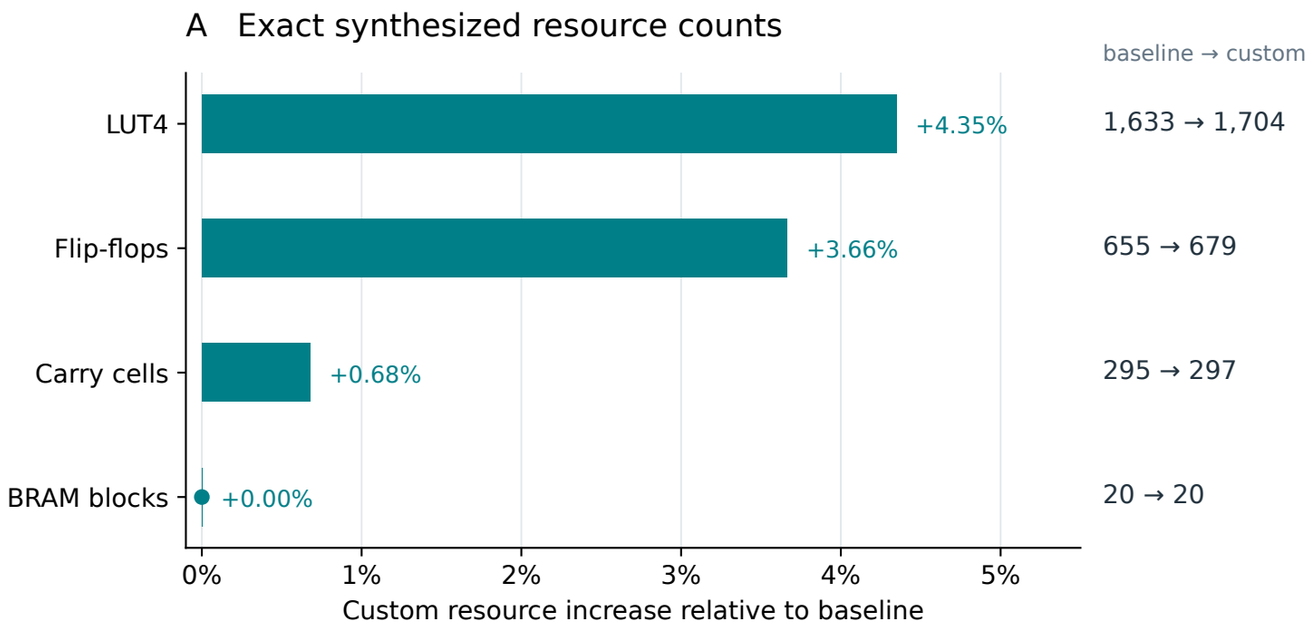
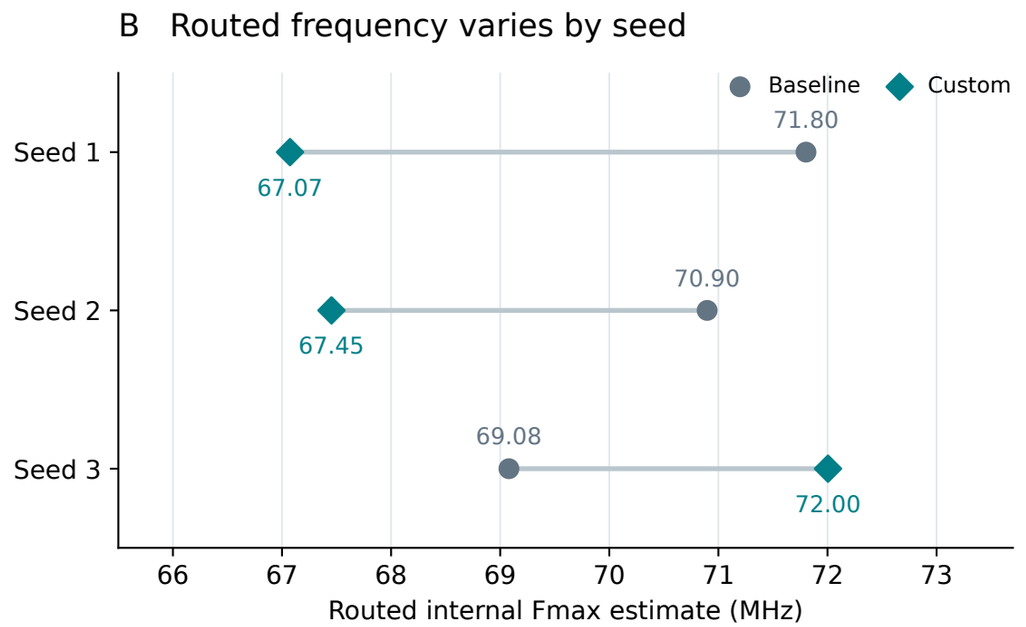
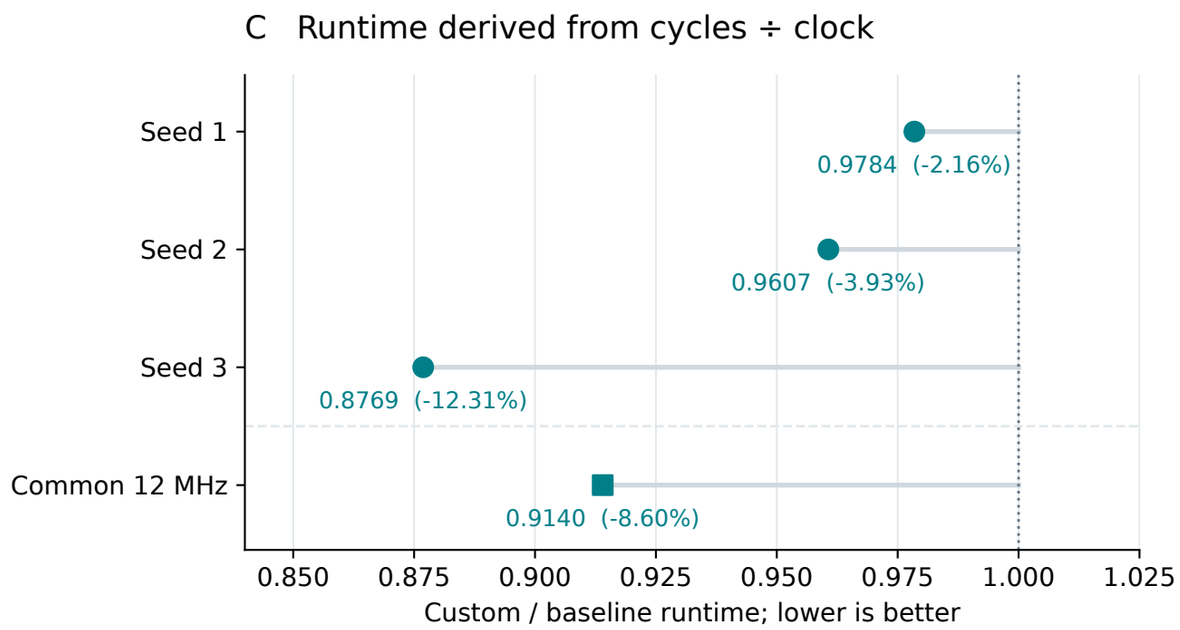
CRC32: integrated hardware cost versus runtime

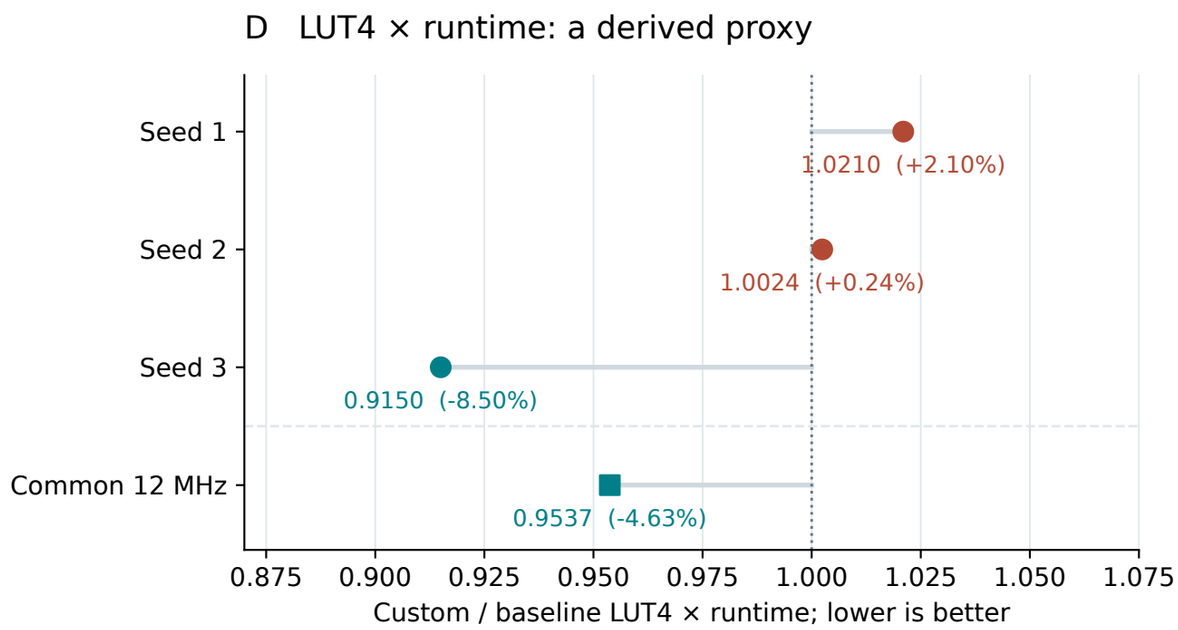
Closed PicoRV32 SoC · iCE40HX8K · full source-modified benchmark · all three matched routing seeds

RTL-simulated cycles: 32,387,382 baseline → 29,602,102 custom (-8.60%) · latency and SoC integration included









Seeds: each design at its own modeled Fmax.
Common clock: 2.699 s baseline → 2.467 s custom.

Product increases in 2 of 3 modeled-Fmax cases.
Proxy uses LUT4 only; not energy or total silicon area.

Cycles from RTL simulation; clocks from nextpnr after routing. No physical board runtime, power or energy measurement.

Three seeds describe placement sensitivity; no confidence intervals. Full precision, formulas and source hashes: figure-data.json.